

QISKIT FALL FEST 2026 @ MPSTME · BUILD A QUANTUM LEARNING APP

Quantum Learn AI

An AI-tutored, Qiskit-powered playground for learning quantum computing by building, seeing, and playing with real circuits.

Tutor

Playground

Games

Team Quantum Learn AI

Round 1 · Project & Presentation Submission

THE PROBLEM

Quantum computing is taught as math before it's felt as intuition

01

Abstract by default

Superposition, entanglement, and interference live in Dirac notation and matrices long before students see them do anything.

02

Static tutorials

Most learning resources are slides or notebooks — read once, run once, no feedback loop back to the learner.

03

No sense of noise

Real quantum hardware is noisy, but classroom examples almost always run on a perfect, idealized simulator.

04

One-size-fits-all pace

Every learner gets the same sequence of topics, whether they're a first-timer or already comfortable with gates.

OUR SOLUTION

One platform, three ways to learn

Every concept is explained, then made touchable — in a real Qiskit circuit, a live simulation, or a game built on quantum randomness.



AI Tutor

Conversational tutor (local LLM, Ollama Qwen3:1.7b) with a rule-based fallback. Tracks per-topic mastery and adapts what it recommends next.



Playground

Click-to-build circuits up to 5 qubits, run on Aer, and see the statevector, Bloch spheres, amplitudes, and QAOA/VQE demos update live.



Games

Guess-the-Qubit, Quantum Coin Flip, and Quantum Dice turn superposition and measurement into quick, replayable challenges.

Concepts the platform makes tangible

H

Superposition

X

Entanglement

Z

Measurement

V

Interference

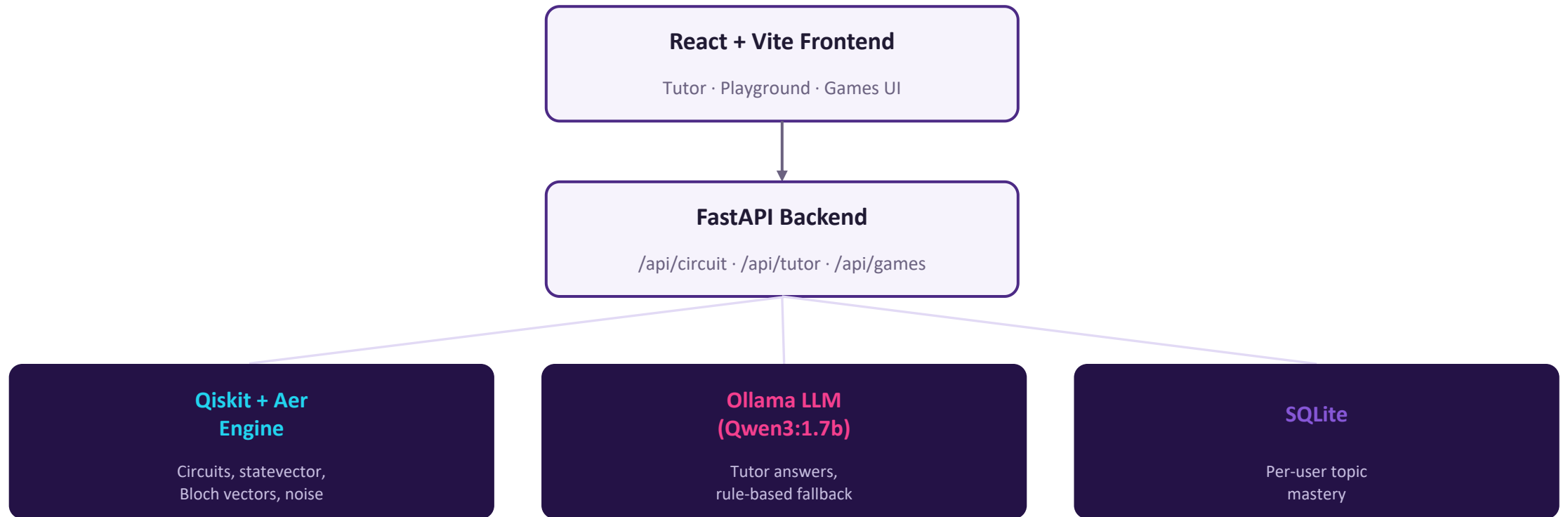
N

Circuit Noise

Σ

Quantum Optimization
(QAOA / VQE)

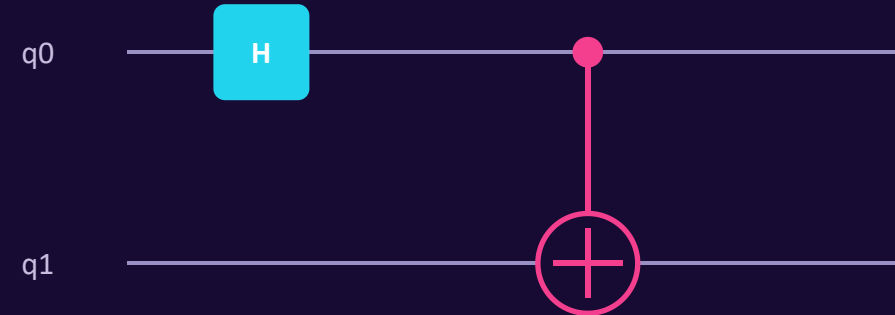
A thin, honest stack



A real circuit builder, not a picture of one

- Click a wire to add a gate: H, X, Y, Z, S, T, CX, CZ, SWAP, RX/RX/RZ — up to 5 qubits.
- Two-qubit gates draw a real connecting line with control / target / swap symbols.
- One-click Bell pair and GHZ presets build genuinely entangled circuits.
- Runs on AerSimulator — ideal or with a depolarizing noise model.

Bell pair preset — $H(q0), CX(q0, q1)$



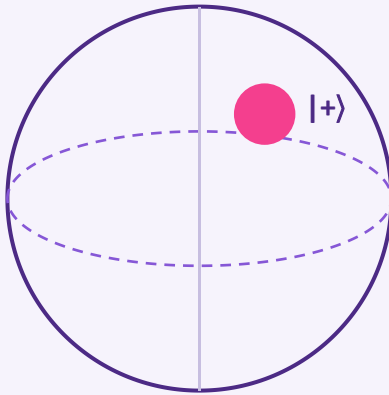
Verified: measurement outcomes are only ever 00 or 11 — never 01/10.

Seeing what a histogram throws away

A measurement histogram only shows probabilities. Phase — the part of a qubit's state that drives interference — disappears. We surface it directly.

3D

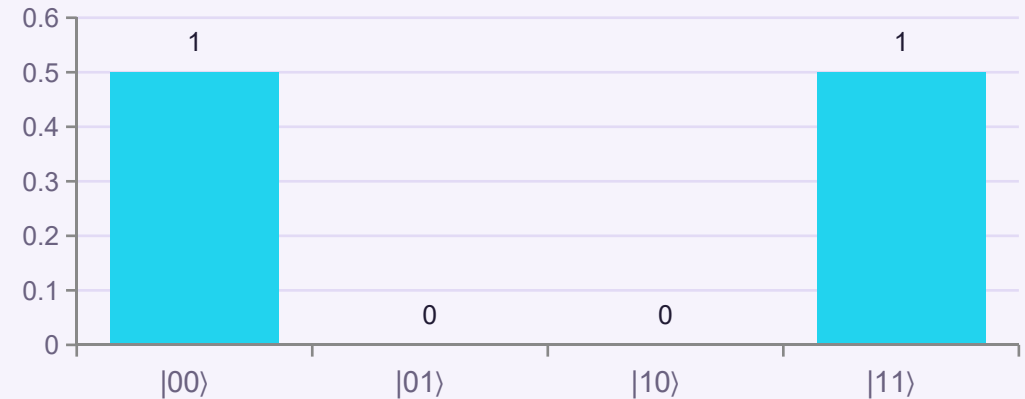
Interactive 3D Bloch Sphere



Rotatable per-qubit sphere (react-three-fiber), reduced from the statevector via partial trace. Verified: $|+\rangle$ renders pointing exactly at the "+" label.

ψ

Amplitude / Phase View



Bell state $|00\rangle + |11\rangle$ shown by magnitude; phase angle colors each bar. Verified against H+S: the $|1\rangle$ component's phase reads exactly 90° .

From ideal qubits to real-world optimization

N

Ideal vs. Noisy Simulation

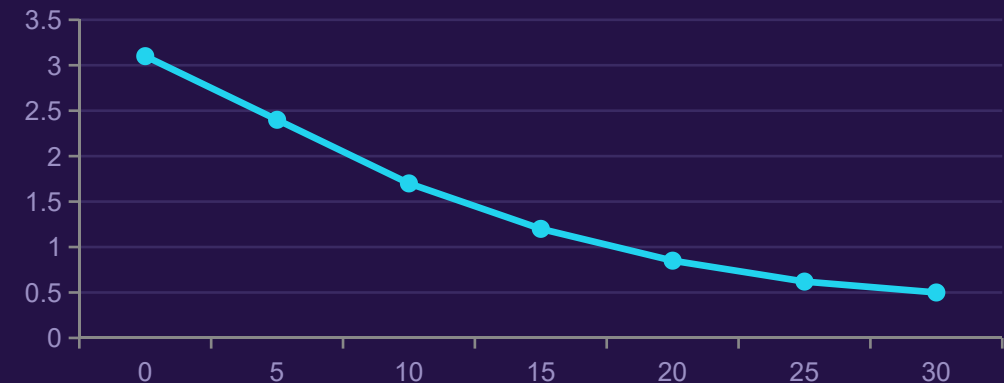
Toggle a depolarizing noise model on any circuit and watch the measurement histogram spread away from the ideal outcome.

This is what makes "quantum computers are noisy" a felt result instead of a warning in a textbook.

Q

QAOA Max-Cut & Toy VQE

Convergence history (qiskit-algorithms)



Optimizer cost trending down across iterations — plotted straight from qiskit-algorithms' convergence history.

Play with quantum randomness, then let the app notice what you know



Guess-the-Qubit

A biased RY rotation sets up the state; you guess the more likely outcome before it measures.



Quantum Coin Flip

A single H gate plus measurement — true 50/50 randomness from superposition, not a PRNG.



Quantum Dice

Superposition across states with rejection sampling gives a fair 6-sided roll from qubits.



Adaptive Mastery Engine

Every tutor question and game round nudges a per-topic mastery score, persisted in SQLite so your "path" survives a restart.

A tutor that still works with zero setup

1

Ask a question

The learner asks about a topic in plain language from the chat UI.

2

Try the local LLM

The backend calls a local Ollama server running Qwen3:1.7b, forwarding recent turns for context and stripping its reasoning traces.

3

Fall back safely

If Ollama isn't running or errors out, the router transparently falls back to curated, rule-based topic explanations — no broken chat.

4

Track engagement

Recognized topics tick mastery up (+0.08); an explicit right/wrong answer moves it further (+0.2 / -0.1).

Verified end-to-end, not just demoed



Bell / GHZ presets

Measurement outcomes are only ever 00/11 (Bell) or 000/111 (GHZ) — confirmed across repeated runs.



Bloch coordinate mapping

$|+\rangle$ renders pointing exactly at the sphere's "+" label — the mapping isn't just visually plausible, it's correct.



Amplitude/phase view

Run against a real H+S circuit, the $|1\rangle$ component's phase reads exactly 90° , matching the expected S-gate rotation.



SQLite persistence

Mastery state written in one backend process was read back correctly in a separate one after a restart.

What's next after Round 1

1

Real accounts

Replace the localStorage-generated user ID with real login and cross-device sync.

2

Drag-and-drop circuits

Let learners drag a placed gate to a new column instead of delete-and-re-add.

3

Leaderboard + new games

Add Quantum Battleship and a superposition-matching game, plus a shared leaderboard.

4

Real VQE molecule

Swap the toy Hamiltonian for an actual small molecule via Qiskit Nature.

5

Full Q-sphere

A 3D, amplitude-weighted Q-sphere for multi-qubit states, complementing the current bar view.

6

Faster Playground load

Code-split the Playground route with React.lazy — the three.js bundle is ~1.5MB today.

Thank you

Quantum Learn AI — built with Qiskit, Aer, FastAPI, React, and a local LLM tutor.

GH

<https://github.com/shelararti/Quantum-Learn-AI>